CCNA LEARNING SERIES

Day 2: OSI Model & TCP/IP Model

Seven layers • Protocols • PDU • Encapsulation

Learning objective
Understand the seven OSI layers, the four-layer TCP/IP model, how data moves through a network, and how these models help with troubleshooting.

OSI layers (top to bottom)

Layer	Name	Main role
7	**Application**	User-facing network services
6	**Presentation**	Data format, encryption, compression
5	**Session**	Manage communication sessions
4	**Transport**	TCP/UDP, ports, end-to-end delivery
3	**Network**	IP addressing and routing
2	**Data Link**	Frames and MAC addresses
1	**Physical**	Bits, cables and signals

A reference guide for beginners, with examples, revision notes and interview questions.

1. What is the OSI Model?

OSI stands for **Open Systems Interconnection**. It is a reference model that explains how two networked systems communicate. It divides the communication process into seven layers, each with a specific responsibility.

Why is it important?
• Helps us understand network communication step by step.
• Gives networking professionals a common language.
• Helps locate possible causes of network problems.
• Explains how protocols and devices work together.

2. The Seven OSI Layers

Layer 7 – Application: Provides network services to applications. Examples include HTTP/HTTPS for web access, DNS for name resolution and SMTP for email.

Layer 6 – Presentation: Handles data representation, translation, encryption/decryption and compression. It helps ensure that data is represented in a format the receiving system can understand.

Layer 5 – Session: Establishes, manages and terminates communication sessions between systems. Session-related functions may be handled by applications and protocols in real TCP/IP implementations.

Layer 4 – Transport: Provides process-to-process communication. It uses port numbers to identify applications and services. TCP and UDP are key transport protocols.

TCP	UDP
Connection-oriented. Provides reliable, ordered delivery and retransmission when needed.	Connectionless. Does not itself guarantee delivery, ordering or retransmission.
Often used when reliable delivery is important.	Often used for DNS queries, voice/video and other suitable applications.

Layer 3 – Network: Handles logical addressing and routing. IP addresses identify source and destination hosts; routers forward packets between networks.

Layer 2 – Data Link: Transfers frames across a local link. Ethernet uses MAC addresses for local delivery. Layer 2 switches learn MAC addresses and forward frames to the appropriate port.

Layer 1 – Physical: Transmits raw bits as electrical, optical or radio signals. It includes cables, connectors, radio transmission and physical signalling.

3. PDU – Protocol Data Unit

A PDU is the name given to data as it is processed at a particular layer. These terms are useful when describing network communication.

Layer	PDU name	Meaning
Layers 5–7	Data	Application information being sent.
Layer 4	Segment (TCP) / Datagram (UDP)	Transport data with port information.
Layer 3	Packet	IP data with source and destination IP addresses.
Layer 2	Frame	Local-link data, including MAC addressing.
Layer 1	Bits	0s and 1s sent as signals.

4. Encapsulation and Decapsulation

Encapsulation happens at the sender. As data moves down the layers, protocol information is added (usually headers; the Data Link layer also adds a trailer). The Physical layer sends the resulting bits.

Decapsulation happens at the receiver. The receiving device processes the data up the layers and interprets or removes the relevant protocol information.

Step	What happens	Resulting PDU
Application / upper layers	Application creates information to send.	Data
Transport	Adds TCP or UDP header, including port information.	Segment / Datagram
Network	Adds IP header with source and destination IP addresses.	Packet
Data Link	Adds local-link frame header and trailer.	Frame
Physical	Transmits the frame as signals.	Bits

Simple example: When you send a message, the sender encapsulates it as it moves down the stack. The receiver decapsulates it as it moves up the stack. The exact headers depend on the protocols being used.

5. What is the TCP/IP Model?

The **TCP/IP model** describes the protocol suite used for communication across modern IP networks. The commonly taught model has four layers.

TCP/IP layer	Main function	Examples
Application	Application services; includes functions represented by OSI Application, Presentation and Session.	HTTP, HTTPS, DNS, SMTP, FTP
Transport	Process-to-process delivery using ports.	TCP, UDP
Internet	Logical addressing and routing between networks.	IPv4, IPv6, ICMP
Network Access	Local network delivery and access to the physical medium.	Ethernet, Wi-Fi

6. OSI Model vs TCP/IP Model

OSI Model	TCP/IP Model
Has seven layers.	Commonly taught with four layers.
A reference model for understanding network functions.	A practical model based on Internet protocols.
Session and Presentation are separate layers.	Their functions are included in Application.
Data Link and Physical are separate layers.	They are combined in Network Access.

7. OSI to TCP/IP Layer Mapping

OSI layers	TCP/IP layer
7 Application, 6 Presentation, 5 Session	Application
4 Transport	Transport
3 Network	Internet
2 Data Link, 1 Physical	Network Access

8. Using the OSI Model for Troubleshooting

The OSI model gives a support engineer a structured way to investigate network problems. You can begin by checking the physical connection and then move upward, depending on the symptoms.

- **Layer 1:** Is the cable connected? Is Wi-Fi signal available? Are link lights on?
- **Layer 2:** Is the correct VLAN configured? Is the switch port up? Is the MAC address learned?
- **Layer 3:** Are the IP address, subnet mask and default gateway correct? Is routing working?
- **Layer 4:** Is the required TCP/UDP port reachable? Could a firewall be blocking it?
- **Layers 5–7:** Is the session working? Does DNS resolve the name? Is the application responding?

9. Interview Questions and Answers

Q1. What is the OSI Model?
A seven-layer reference model that explains how data communication happens between networked systems.

Q2. Name all seven OSI layers.
Application, Presentation, Session, Transport, Network, Data Link and Physical.

Q3. Which layer is responsible for routing?
Layer 3, the Network layer.

Q4. Which layer uses MAC addresses?
Layer 2, the Data Link layer, uses MAC addresses for local-link delivery.

Q5. Which layer uses port numbers?
Layer 4, the Transport layer.

Q6. What is the difference between TCP and UDP?
TCP provides connection-oriented, reliable and ordered delivery. UDP is connectionless and does not itself guarantee delivery or ordering.

Q7. What is a PDU?
Protocol Data Unit: the name for data as it is processed at a particular layer.

Q8. What is encapsulation?
Adding protocol information as data moves down the layers at the sender.

Q9. What is decapsulation?
Processing protocol information as data moves up the layers at the receiver.

Q10. How many layers are in the TCP/IP model?
The commonly taught TCP/IP model has four layers.

Q11. Which OSI layers map to TCP/IP Application?
OSI Layers 5, 6 and 7: Session, Presentation and Application.

Q12. At which layer does a traditional Layer 2 switch operate?
Layer 2, the Data Link layer.

10. Quick Revision

Remember	Answer
OSI layers	7 Application, 6 Presentation, 5 Session, 4 Transport, 3 Network, 2 Data Link, 1 Physical
TCP/IP layers	Application, Transport, Internet, Network Access
Layer 4 PDU	TCP segment / UDP datagram
Layer 3 PDU	Packet
Layer 2 PDU	Frame
Layer 1 PDU	Bits
Encapsulation	Sender: data moves down; protocol information is added.
Decapsulation	Receiver: data moves up; protocol information is processed.

Study tip: Practise explaining each OSI layer in one simple sentence. Then take an example—opening a website—and explain how the layers help communication happen.

End of Day 2 | Next topic: IPv4 Addressing and Subnetting

Note: This guide uses the commonly taught four-layer TCP/IP model. Some learning materials use a five-layer model by separating Network Access into Data Link and Physical. Protocol-to-layer mappings are conceptual; real protocol stacks do not always fit perfectly into one OSI layer.